

Alejandro Vaz

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Computer Science student specializing on tokenizing, parsing, and performance.

Professional Experience

Rust SmallVec **03/26 - Present**
Open Source Contribution

Contributed to SmallVec performance on v2 branch.

- Solved the performance regressions v2 introduced.
- Improved extend performance by +140%.
- Improved push performance by +30%.
- Improved pushpop performance by +40%.
- PR currently under review.

Software Language Engineering 2026 Conference **02/26 - 04/26**
Independent researcher.

Submitted paper of Mathsys' disambiguation algorithm.

Education

Colegio Mater Immaculata / IES Lope de Vega **09/24 - Present**
Bachillerato Tecnológico - 10.0/10 with three honors

- Founded the school's programming club which reached 34 members across five grade levels.

Key Skills

- Rust.
- Python.
- Performance engineering.
- Compiler tooling.
- Effective communication.
- Team work.

Projects

Mathsys **07/25 - Present**

- Domain-Specific Language (DSL).
- Hand-rolled parser (400M tok/s, ASCII, zero-copy, byte scanner).
- Hand-rolled Earley parser (75ms per 4k token chunk, memoization).
- Disambiguation algorithm.

CSScribe **11/24 - 03/25**

- Markup Language for writing research papers.
- Compiling pipeline was one-step and used regular expression replacement.
- Written on Python.
- Compiled to Markdown.